Kuta Software - Infinite Calculus

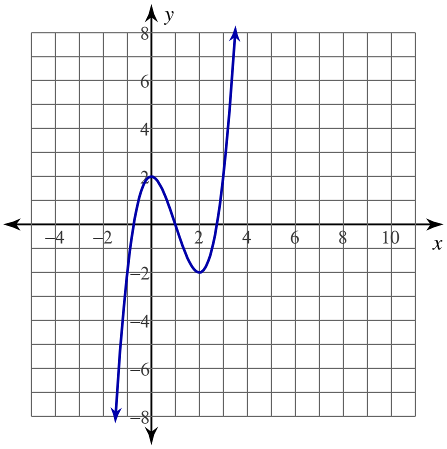
Name________________________________

# Tangent Lines

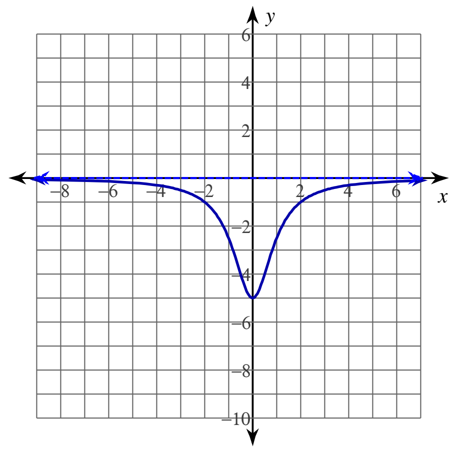
Date________________ Period____

**For each problem, find the equation of the line tangent to the function at the given point. Your answer should be in slope-intercept form.**

1) $y = x^3 - 3x^2 + 2$ at $(3, 2)$



2) $y = -\dfrac{5}{x^2 + 1}$ at $\left(-1, -\dfrac{5}{2}\right)$



3) $y = x^3 - 2x^2 + 2$ at $(2, 2)$

4) $y = -\dfrac{3}{x^2 - 25}$ at $\left(-4, \dfrac{1}{3}\right)$

5) $y = -\dfrac{3}{x^2 - 4}$ at $(1, 1)$

6) $y = (5x + 5)^{\frac{1}{2}}$ at $(4, 5)$

7) $y = \ln(-x)$ at $(-2, \ln 2)$

8) $y = -2\tan(x)$ at $(-\pi, 0)$